Here's your **MCQ-based quiz (without answers)** for B.Tech students, covering all requested topics in a structured and practical way.

🧠 Azure Cognitive Services & AI Deployment Quiz (MCQs)

🔹 SECTION 1: Cognitive Services Overview & Architecture

Q1.

Which Azure service is best suited for extracting structured data from invoices and forms?
A. Speech Service
B. Language Service
C. Document Intelligence
D. Computer Vision

Q2.

In Azure Cognitive Services architecture, what is the primary role of an endpoint?
A. Store training data
B. Provide access to AI models via API
C. Manage user authentication
D. Monitor application logs

Q3.

Which component is responsible for converting speech into text?
A. Translator
B. Speech Service
C. Vision Studio
D. Bot Framework

Q4.

Which architecture pattern is commonly used when multiple AI services are combined?
A. Monolithic architecture
B. Microservices architecture
C. Peer-to-peer
D. Standalone system

Q5.

Which Azure service helps detect objects, faces, and text within images?
A. Language Service
B. Computer Vision
C. Speech Service
D. Form Recognizer

◆ SECTION 2: Service Categories & Use Cases

Q6.

Which service is used for sentiment analysis?
A. Computer Vision
B. Language Service
C. Speech Service
D. Translator

Q7.

Which use case best fits the Speech Service?
A. Image classification
B. Real-time voice assistant
C. Document parsing
D. Language translation

Q8.

Which service is most appropriate for building a multilingual chatbot?
A. Translator + Language Service
B. Computer Vision
C. Face API
D. Form Recognizer

Q9.

Which category does OCR (Optical Character Recognition) belong to?
A. Speech AI
B. Vision AI
C. Language AI
D. Decision AI

Q10.

Which scenario requires combining multiple Cognitive Services?
A. Converting text to speech
B. Detecting faces in an image
C. Analyzing video with audio and subtitles
D. Translating text

◆ SECTION 3: API Integration Patterns

Q11.

Which protocol is most commonly used to call Azure Cognitive Services?
A. FTP
B. REST
C. SMTP
D. TCP

Q12.

What is required to authenticate API calls to Azure Cognitive Services?
A. Username and password
B. API key or token
C. Database connection string
D. IP address

Q13.

Which integration method provides pre-built functions and easier coding?
A. REST API
B. SDK
C. CLI
D. Portal

Q14.

Which pattern is best for handling large-scale asynchronous processing?
A. Blocking calls
B. Batch processing with queues
C. Direct synchronous calls
D. Manual processing

Q15.

What is the purpose of retry logic in API integration?
A. Improve UI design
B. Handle temporary failures
C. Reduce API cost
D. Increase response time

◆ SECTION 4: Deployment Strategies

Q16.

Which deployment model is best for applications requiring global scalability?
A. On-premises

B. Cloud-based deployment
C. Local server
D. Standalone system

Q17.

Which deployment is ideal for low-latency and offline scenarios?
A. Cloud deployment
B. Edge deployment
C. Hybrid deployment
D. SaaS deployment

Q18.

What is a key advantage of container-based deployment?
A. Requires no internet
B. Easy portability across environments
C. No need for security
D. No configuration required

Q19.

Which Azure feature helps automatically scale applications?
A. Virtual Machines
B. Auto-scaling
C. Storage Accounts
D. Resource Groups

Q20.

Which strategy helps reduce cost in AI deployments?
A. Always using highest pricing tier
B. Scaling resources based on demand
C. Avoiding monitoring
D. Using only on-premises systems

◆ SECTION 5: Identity & Access Management (IAM)

Q21.

Which service is used for identity management in Azure?
A. Azure DevOps
B. Azure Active Directory
C. Azure Storage
D. Azure Monitor

Q22.

Which authentication method is more secure?
A. API keys
B. Managed Identity
C. Password-based login
D. Anonymous access

Q23.

What is the main purpose of role-based access control (RBAC)?
A. Improve UI
B. Control user permissions
C. Increase speed
D. Store data

Q24.

What should you do if an API key is exposed publicly?
A. Ignore it
B. Regenerate the key immediately
C. Share it with team
D. Disable the application

◆ SECTION 6: Responsible AI, Security & Compliance

Q25.

Which principle ensures AI systems treat all users fairly?
A. Reliability
B. Fairness
C. Transparency
D. Privacy

Q26.

Which technique helps reduce harmful AI outputs?
A. Increasing model size
B. Guardrails and content filtering
C. Removing APIs
D. Disabling authentication

Q27.

What is the purpose of data encryption?
A. Improve speed
B. Protect sensitive data
C. Reduce storage
D. Simplify UI

Q28.

Which regulation focuses on data privacy?
A. HTTP
B. GDPR
C. TCP
D. HTML

Q29.

What is bias in AI systems?
A. Faster processing
B. Unfair or skewed outcomes

C. High accuracy
D. Secure processing

Q30.

Which practice improves AI transparency?
A. Hiding model logic
B. Providing explanations for predictions
C. Ignoring outputs
D. Reducing data